

AI for Chemistry

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Introduction

Predicting chemical reactivity is a central challenge in computational chemistry, as the electrophilic and nucleophilic character of molecular sites strongly influences reaction mechanisms, selectivity, and synthesis design. Traditionally, such properties are estimated through quantum chemistry calculations, which can become computationally expensive for large molecular libraries.

Among the most widely used quantitative reactivity scales are the Mayr electrophilicity and nucleophilicity parameters, which relate reaction kinetics to the intrinsic reactivity of electrophiles and nucleophiles. In particular, the methyl anion affinity (MAA) and methyl cation affinity (MCA) provide computational descriptors directly linked to electrophilic and nucleophilic behavior at the atomic level.

The objective of this project is therefore to develop machine learning models capable of predicting site-specific MAA and MCA values directly from molecular structure. Unlike standard molecular property prediction tasks, this problem requires distinguishing between multiple reactive sites within the same molecule, making local structural information particularly important.

To address this challenge, several approaches were investigated, ranging from tree-based models using molecular fingerprints to graph neural networks operating directly on molecular graphs. Particular attention was also paid to dataset splitting strategies and leakage prevention in order to ensure chemically meaningful evaluation of model generalization.

1 Description of the Predicted Parameters

To quantify electrophilicity and nucleophilicity, Mayr’s equation can be used :

$$\log k_{20^{\circ}\text{C}} = s_N \cdot s_E (N + E) \quad (1)$$

where $k_{20^{\circ}\text{C}}$ is the rate constant at 20°C, and s_N , s_E , N , and E are experimentally determined parameters representing the nucleophile-specific sensitivity factor, the electrophile-specific sensitivity factor, the nucleophilicity, and the electrophilicity, respectively. Since $s_E \approx 1$ for most electrophiles, Equation 1 simplifies considerably. Furthermore, it has been shown that the parameter E is linearly proportional to the methyl anion affinity (MAA) [2], while the methyl cation affinity (MCA) is linearly proportional to the product $s_N \cdot N$ [3].

The MCA and MAA values represent the reaction energy of the following processes :



The models developed in this work were trained to predict the MCA and MAA values for every atom in a given molecule. To this end, two datasets from the Jensen group [1] were used : one for MCA values, comprising 650 857 unique atomic sites from 47 921 unique molecules, and one for MAA values, comprising 534 119 unique atomic sites from 47 440 unique molecules. Both datasets provide MCA and MAA values (in kJ mol^{-1}) for the nucleophilic and electrophilic sites of each molecule, computed using the composite $r^2\text{SCAN-3c}$ method in DMSO [4].

2 Tree based models

a) Choosing the Model

As an initial baseline, tree-based models such as XGBoost and RandomForest were evaluated. To define which model with which fingerprints were best suited for the determination of MAA values, many

combinations were tested. They are all summarized in the table below.

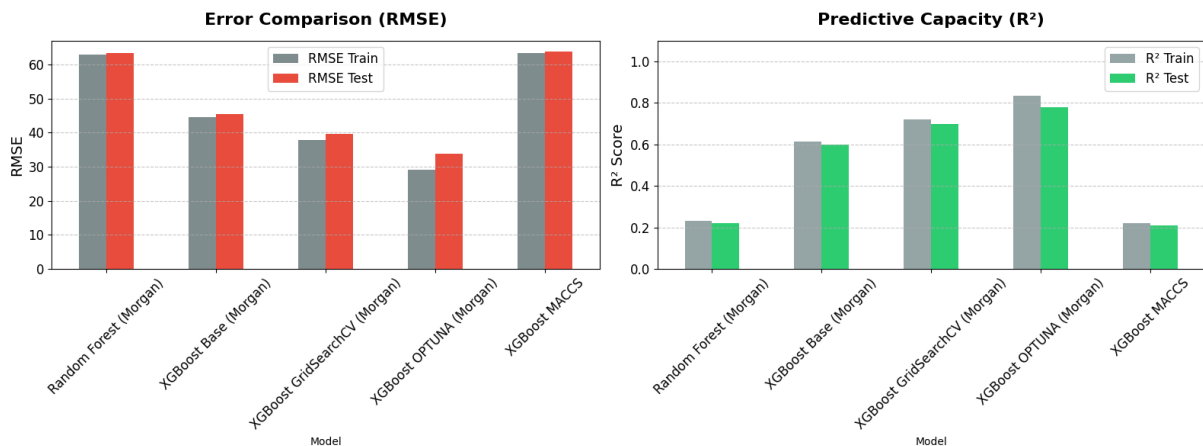


FIGURE 1 – Tree based models

First, baseline XGBoost and Random Forest models were compared using default parameters. XGBoost outperformed Random Forest in both RMSE and R^2 , establishing it as the superior core algorithm for this task.

b) Selecting the Features

Next, two featurization strategies were evaluated to encode the chemical structures : MACCS structural keys and site-specific Morgan circular fingerprints. MACCS keys rely on a fixed 166-bit vector tracking predefined fragments. Because they are strictly global, they lack the spatial resolution required to differentiate between distinct reactive sites within the same molecule, resulting in a poor R^2 of 0.21 when paired with XGBoost.

Conversely, the Morgan fingerprinting approach maps molecular topologies dynamically based on a specific atom-neighborhood radius ($r = 2$). To capture both macroscopic molecular properties and the microscopic environment of the target electrophilic center, a dual-component function was designed. It concatenates a global molecular fingerprint (radius = 2, 2048 bits) with a local atom-centered fingerprint (radius = 2, 2048 bits). This site-specific 4096-bit representation successfully preserved crucial spatial context, significantly boosting the model’s performance to an R^2 of 0.60.

c) Defining the Optimization

Finally, two distinct hyperparameter optimization strategies were compared : GridSearchCV and the Optuna Bayesian framework. The Optuna-optimized model achieved a superior R^2 score, nearly reaching 0.80. This performance increase was expected, as Optuna employs a more advanced sequential search paradigm. Unlike the exhaustive, brute-force approach of GridSearchCV—which evaluates pre-defined parameter combinations independently—Optuna utilizes past evaluation history to probabilistically guide subsequent trials toward the most promising regions of the hyperparameter space.

d) Improving the XGBoost, Morgan fingerprints model

However, upon evaluating the model on various molecular schemas, a significant number of outliers persisted. To address this issue, it was decided not to train the model directly on the initial MAA values provided in the dataset. Instead, a “Master model” was developed, based exclusively on the elec_GCS_3_cm5

values—topological and electronic descriptors computed via Density Functional Theory (DFT) and the Charge Model 5 (CM5) method.

These descriptors, available for the entire dataset, allowed the Master model to calculate new smoothed MAA targets with an excellent coefficient of determination ($R^2 \approx 0.98$). Because these new target values were derived from a straightforward regression architecture, the anomalous outliers were successfully eliminated.

Subsequently, an XGBoost regressor was trained on these newly calculated targets, and hyperparameter optimization was performed via Optuna. This final optimized model achieved an R^2 of 0.8274 on the Master model’s predictions, and an R^2 of 0.8048 on the true baseline MAA values, notably improving upon the prior baseline of 0.778. Concurrently, the RMSE decreased from 33.83 to 31.72 kcal·mol⁻¹. While this represents a moderate quantitative gain, this methodology proved to be a robust and worthwhile refinement strategy.

3 Graph Neural Network model

a) Graph Attention Network for Electrophilicity and Nucleophilicity Prediction

Each molecule was encoded as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes represent heavy atoms and edges represent covalent bonds. Atom features consist of a 12-dimensional vector encoding atomic number, degree, formal charge, number of hydrogens, aromaticity, ring membership, hybridisation state (one-hot, 5 types), and a binary flag indicating the electrophilic site of interest. Each bond is described by a 6-dimensional vector encoding bond type (one-hot, 4 types), conjugation, and ring membership.

The model consists of three successive Graph Attention convolution layers with 4, 4, and 1 attention heads, producing representations of dimension 256, 512, and 256, respectively. Self-loops were disabled to accommodate edge features. After the final convolution, the representation of the target atom is extracted via the site mask and passed through a two-layer feed-forward head ($256 \rightarrow 128 \rightarrow 1$) to produce the predicted MAA or MCA value.

The dataset was split at the molecule level (80/10/10) using a fixed random permutation (`numpy.default_rng(42)`) on canonical SMILES, with an explicit zero-overlap assertion between splits. Target values were standardised using statistics computed on the training set only.

Training used the AdamW optimiser ($\text{lr} = 10^{-3}$, weight decay = 10^{-5}) with gradient clipping (norm = 5.0), a `ReduceLROnPlateau` scheduler (factor 0.5, patience 5), and early stopping (patience 15, max 200 epochs). The Huber loss was used as the training objective. Model selection was based on the lowest validation loss, and performance was reported on the held-out test set in terms of R^2 , MAE, and RMSE.

a).1 Evaluation of the models

The GAT model trained to predict MCA values, achieved :

$$\text{RMSE} = 25.22 \text{ kcal/mol}, \quad \text{MAE} = 17.50 \text{ kcal/mol}, \quad R^2 = 0.878$$

The parity plot shows a strong linear correlation over the full range, though a cluster near 320 kcal/mol suggests mild difficulty with specific molecular environments 2a.

The GAT model for MAA prediction yielded :

$$\text{RMSE} = 30.24 \text{ kcal/mol}, \quad \text{MAE} = 21.29 \text{ kcal/mol}, \quad R^2 = 0.821$$

The broader scatter reflects the greater difficulty of electrophilicity prediction.

b) Site-aware GINE model for electrophilicity prediction

A second graph neural network architecture based on GINE convolutions was developed to improve site-level electrophilicity prediction. Unlike the previous attention-based model, GINE convolutions explicitly incorporate bond features during message passing, making them particularly suitable for chemistry applications.

Each molecule was represented as a graph where atoms correspond to nodes and covalent bonds to edges. Atom features included atomic number, degree, aromaticity, hybridisation, formal charge, and a binary flag identifying the electrophilic site of interest. Bond features encoded bond type, conjugation, and ring membership. The resulting model, referred to as *SiteGINE*, consists of five GINEConv layers with residual connections and LayerNorm normalization, followed by global mean/max/sum pooling and a three-layer regression head.

b).1 Correction of data leakage

A major methodological issue was identified during development. Since the ESNUEL MAA dataset contains 534 119 site-level rows but only 47 440 unique molecules (approximately 11.3 sites per molecule), a naive row-level random split causes the same molecule to appear simultaneously in train, validation, and test sets through different reactive sites. This introduces direct data leakage and artificially inflates performance.

To correct this issue, all experiments reported below use strict molecule-level splitting. Canonical SMILES were first partitioned into train/validation/test subsets before assigning site-level rows, and an explicit overlap assertion verified zero SMILES overlap between splits.

b).2 Evaluation under different splitting strategies

Three increasingly demanding splitting strategies were investigated.

The first experiment used a random molecule-level split on filtered MAA values in the chemically reasonable range [0, 400] kcal/mol. SiteGINE achieved :

$$\text{RMSE} = 14.46 \text{ kcal/mol}, \quad \text{MAE} = 11.00 \text{ kcal/mol}, \quad R^2 = 0.937$$

A second experiment used a Bemis–Murcko scaffold split, ensuring that molecular families absent from training appeared exclusively in the test set. Despite this much stricter criterion, performance remained almost unchanged :

$$\text{RMSE} = 14.86 \text{ kcal/mol}, \quad \text{MAE} = 11.30 \text{ kcal/mol}, \quad R^2 = 0.934$$

This surprisingly small degradation suggests that the model learns transferable local electronic environments rather than simply memorizing molecular scaffolds.

Finally, a chemically motivated MAA-range split was introduced. Molecules were ranked according to their maximum site MAA value :

- train : max MAA < 228 kcal/mol,
- validation : 228–252 kcal/mol,
- test : > 252 kcal/mol.

This configuration forces the model to extrapolate toward highly electrophilic molecules poorly represented during training. Under these conditions, performance decreased substantially :

$$\text{RMSE} = 34.08 \text{ kcal/mol}, \quad \text{MAE} = 26.25 \text{ kcal/mol}, \quad R^2 = 0.766$$

The parity plot (Appendix A) reveals a systematic underestimation of very high MAA values, characteristic of regression-to-the-mean effects during extrapolation beyond the training distribution.

An additional experiment was performed without filtering extreme MAA values. Although this configuration produced the highest reported R^2 (0.950), RMSE and MAE were both slightly worse than in the filtered random split. This result highlights the sensitivity of R^2 to target variance and confirms that RMSE and MAE constitute more reliable metrics when comparing models across different target distributions.

Overall, the SiteGINE architecture demonstrates excellent interpolation and scaffold-level generalization capabilities within the moderate electrophilicity regime, while simultaneously revealing a clear limitation when extrapolating toward strongly electrophilic molecules underrepresented in the training distribution.

b).3 Extension to nucleophilicity prediction

The SiteGINE architecture was also applied to the MCA nucleophilicity dataset, comprising 650 857 site-level rows from 47 921 unique molecules, using the same architecture, training procedure, and molecule-level splitting strategy as for the MAA experiments.

The MCA distribution differs substantially from MAA : values are approximately centered around 315 kcal mol⁻¹ with a standard deviation of 73 kcal mol⁻¹. Consequently, the filtering interval was adjusted to [0, 600] kcal mol⁻¹, and a slightly larger learning rate (lr = 3 × 10⁻³) was adopted to compensate for the weaker relative gradient signal under the MSE objective.

Under a random molecule-level split, SiteGINE (cf. Figure 3 Annex) :

$$\text{RMSE} = 13.90 \text{ kcal mol}^{-1}, \quad \text{MAE} = 10.27 \text{ kcal mol}^{-1}, \quad R^2 = 0.963.$$

This performance marginally surpasses the electrophilicity results. Several factors likely contribute to this improvement : the larger training set (approximately 520 000 versus 393 000 training graphs), the more homogeneous nature of nucleophilic reactive sites (primarily lone-pair donors and π -systems), and the more symmetric target distribution, all of which facilitate generalization.

Furthermore, the normalized error remains lower for MCA prediction :

$$\frac{\text{RMSE}}{\sigma} = 0.191 \quad \text{vs} \quad 0.250 \text{ for MAA},$$

confirming that the improvement is not solely an artifact of the larger target variance reflected in the higher R^2 value.

4 Conclusion

In this work, several machine learning approaches were developed to predict electrophilicity and nucleophilicity descriptors from molecular structures. Tree-based methods demonstrated that site-specific Morgan fingerprints combined with XGBoost and Optuna optimization significantly improved predictive performance compared to simpler descriptors and baseline models. The introduction of a Master model based on DFT-derived descriptors further reduced outliers and enhanced robustness. In parallel, graph neural network architectures, especially the SiteGINE model, achieved excellent interpolation and scaffold generalization performances, reaching R^2 values above 0.93 under molecule-level splitting strategies, and extending to nucleophilicity prediction with $R^2 = 0.963$ on MCA values, confirming the generality of the approach across both reactivity descriptors. However, extrapolation toward highly electrophilic compounds remained challenging, highlighting the limitations imposed by training data distribution. Overall, this study confirms the strong potential of modern AI methods for reactivity prediction in computational chemistry while emphasizing the importance of rigorous dataset splitting and chemically meaningful evaluation protocols.

Bibliography

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- [3] T. KRAMER, M. GINDELE et H. MAYR. "Methyl Cation Affinities : A Quantitative Scale for the Nucleophilicity of Carbon-Centered Nucleophiles". In : *The Journal of Organic Chemistry* (2020). DOI : 10.1021/acs.joc.0c02327.
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A Parity Plots

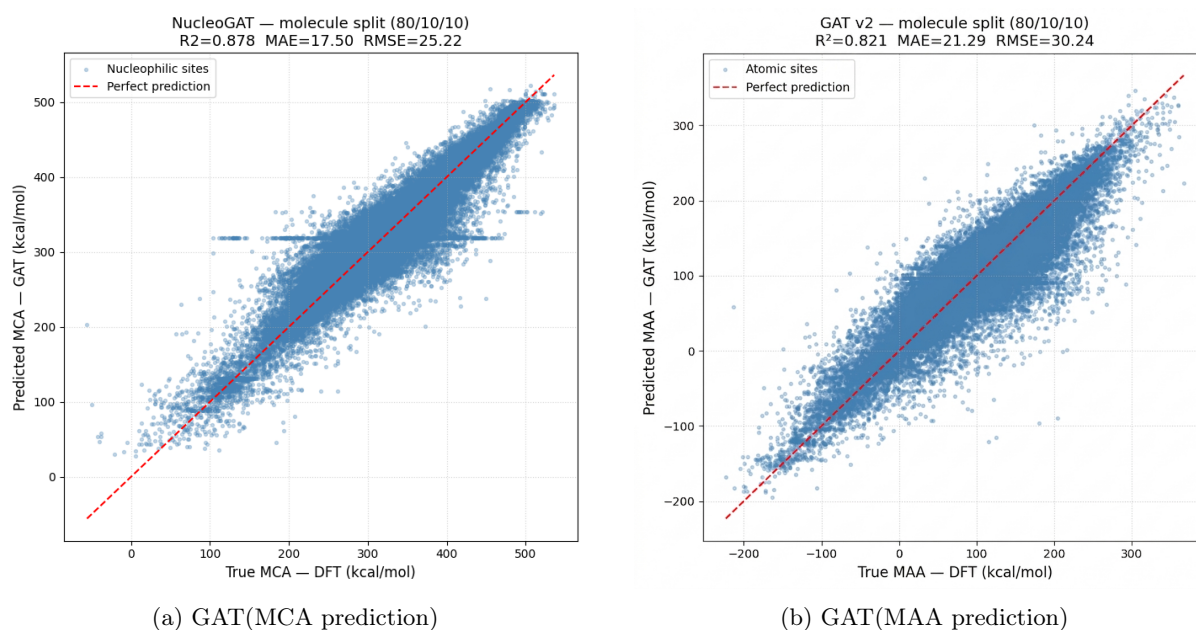
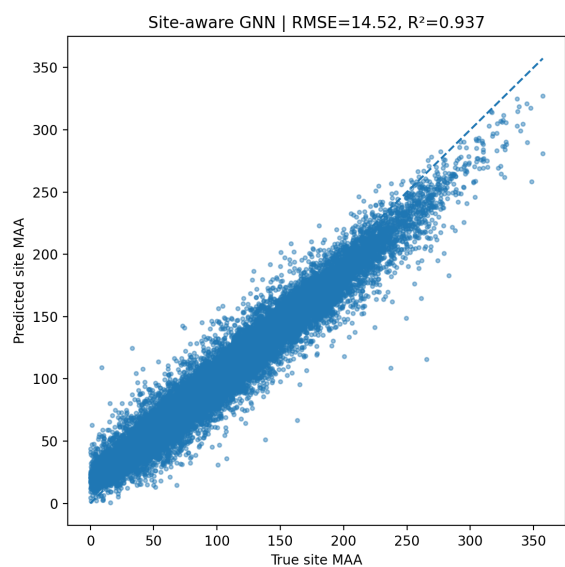
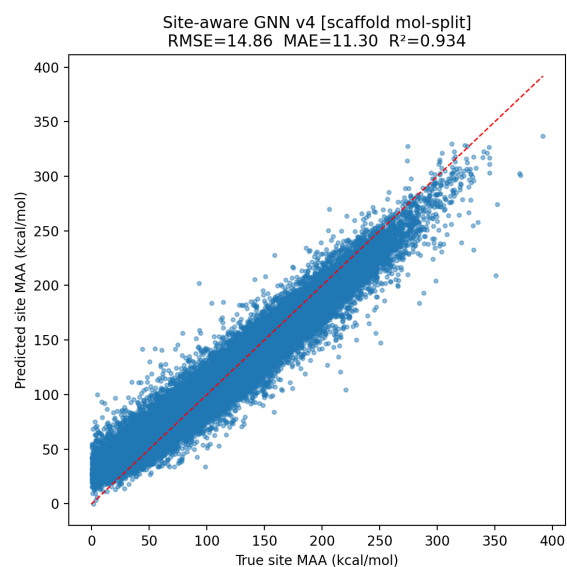


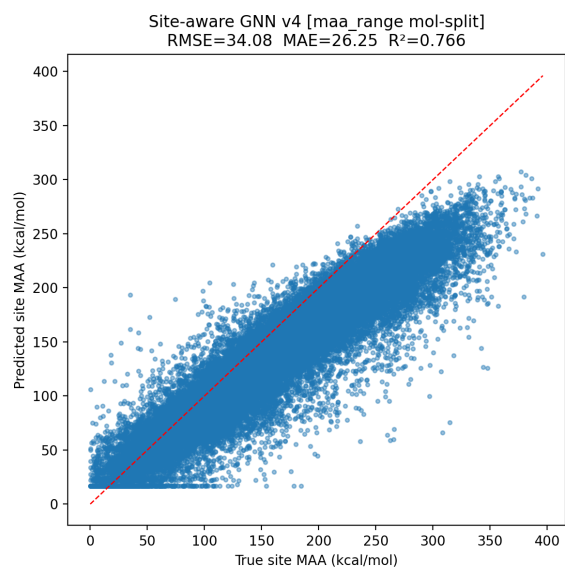
FIGURE 2 – Regression parity plots obtained under the 80/10/10 random molecule split for : (a) NucleoGAT model ($R^2 = 0.878$), and (b) GAT model ($R^2 = 0.821$). The dashed red line indicates the ideal $y = x$ prediction.



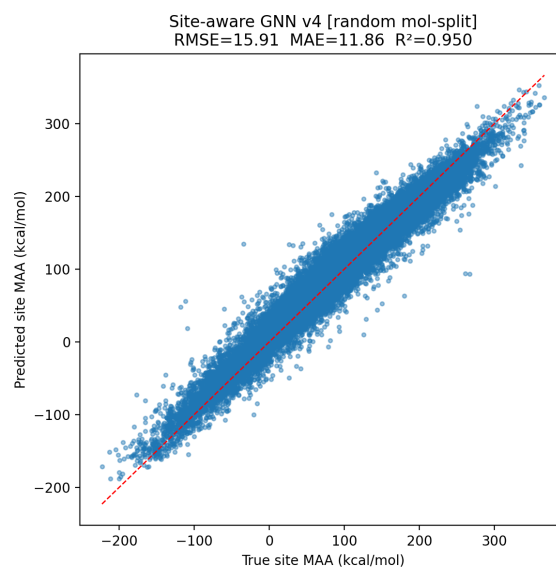
(a) Random molecule split



(b) Scaffold split



(c) MAA-range split



(d) No MAA filtering

FIGURE 3 – Parity plots obtained under the different splitting and filtering strategies investigated for the SiteGINE model.

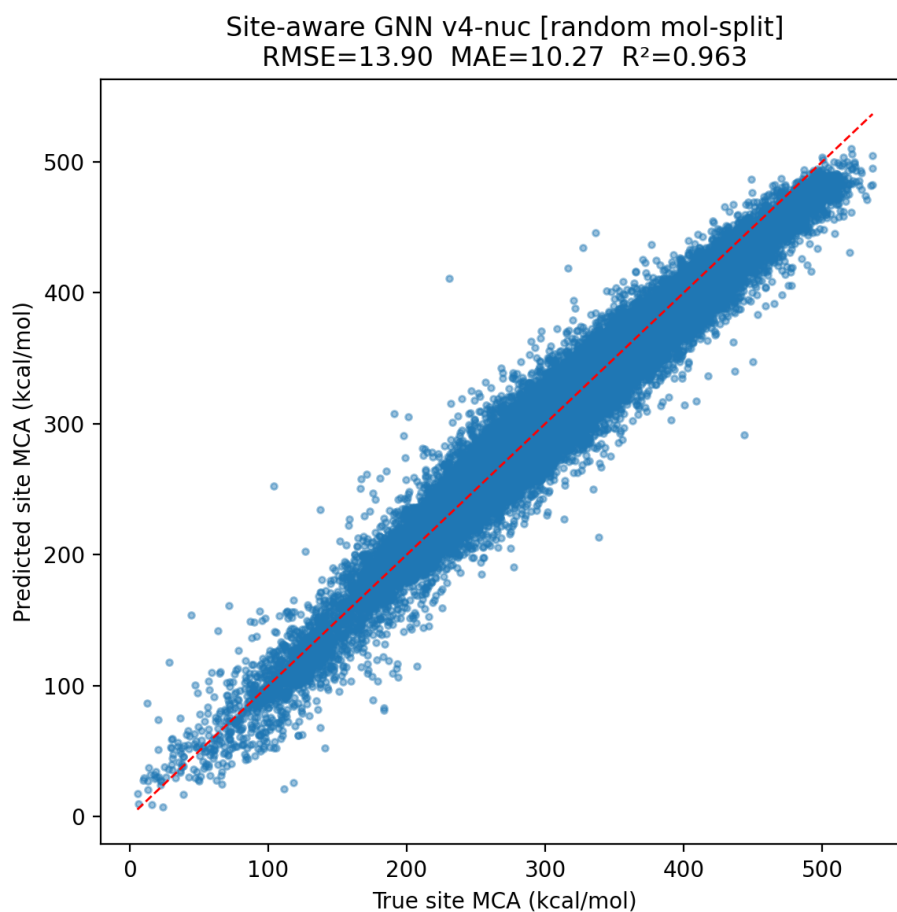


FIGURE 4 – Parity plot obtained for MCA nucleophilicity prediction using the SiteGINE model under a random molecule-level split.